Industrial Internship Report on

”Banking Information System”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was “**Banking Information System using Python**”, which simulates basic banking operations like account creation , deposit , withdrawal , and balance checking. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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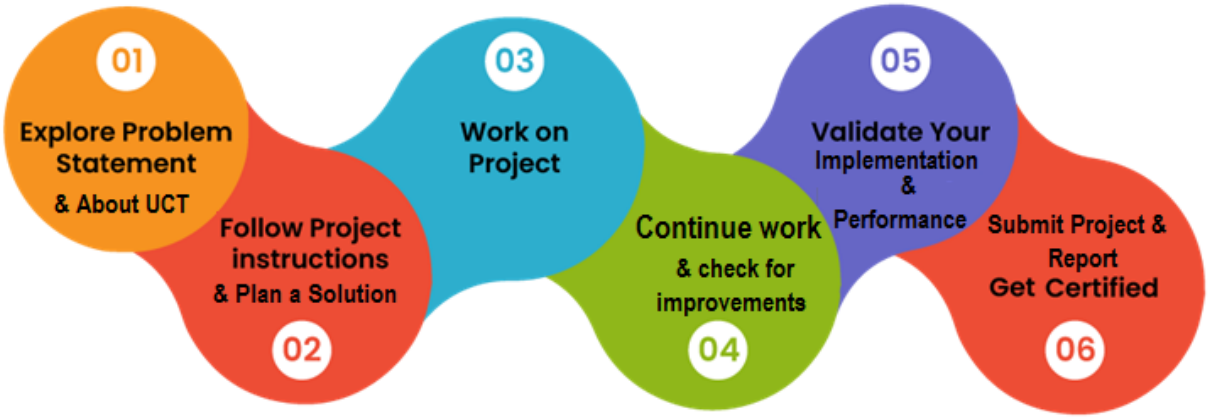
1 Preface

This report summarizes the work completed during the 6-week internship.

Internships are important for career development as they bridge the gap between theoretical knowledge and practical implementation. My project focused on developing a Banking Information System that performs core banking operations.

The internship was well-structured with learning modules, assignments, and quizzes. I learned Python fundamentals and applied them to build this project.

I would like to thank **Upskill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd,** along with my mentors, for their support and guidance.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

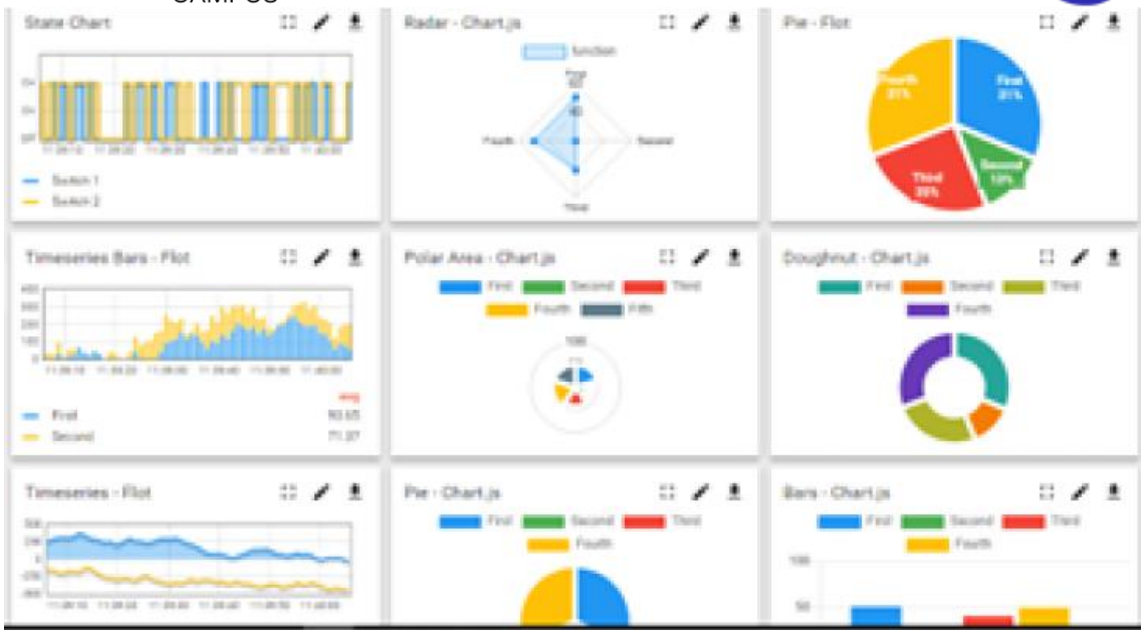
For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

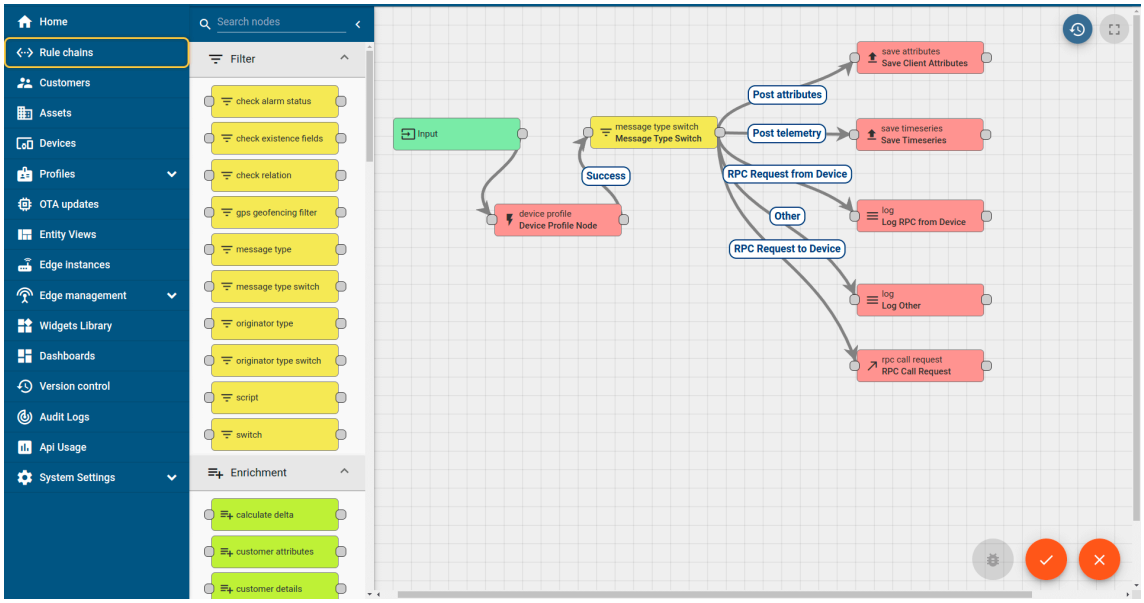
i. UCT IoT Platform (UCT Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.

- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.

- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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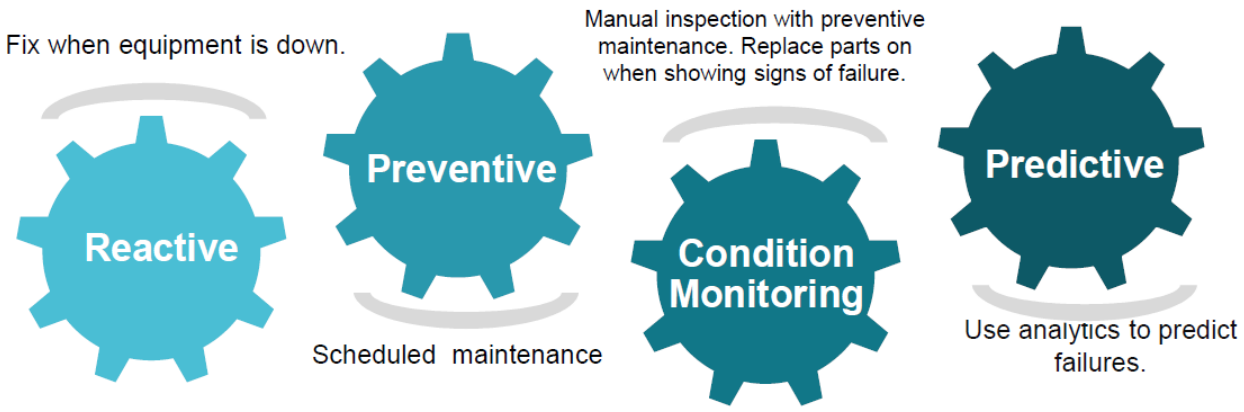
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

3..

3.1 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems .
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

3.2 Reference

[1] Python Official Documentation

[2] Online Tutorials and Learning Resources

[3] Internship Study Material

2.6 Glossary

Terms	Acronym
DBMS	Database Management System
GUI	Graphical User Interface
API	Application Programming Interface

4 Problem Statement

The objective of this project is to develop a **Banking Information System** that allows users to perform basic banking operations efficiently.

The system should:

- Create new bank accounts
- Deposit money
- Withdraw money

- Check account balance

Store and manage user data

5 Existing and Proposed solution

Existing Solution

- Manual record keeping is slow
- High risk of errors
- Time-consuming processes

Proposed Solution

- Develop a Python-based automated system
- Store user details digitally
- Perform transactions efficiently

Value Addition

- Faster operations
- Reduced errors
- User-friendly system

5.1 Code submission (Github link)

https://github.com/Sonal87-coder/upskillcampus/blob/main/BankingInformationSystem.py

5.2 Report submission (Github link) :

https://github.com/Sonal87-coder/upskillcampus/blob/main/BankingInformationSystem_Sonal_USC_UCT.pdf.

6 Proposed Design/ Model

Given more details about design flow of your solution. This is applicable for all domains. DS/ML Students can cover it after they have their algorithm implementation. There is always a start, intermediate stages and then final outcome.

6.1 High Level Diagram (if applicable)

User → Banking System → Output

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

6.2 Low Level Diagram (if applicable)

Input : User details

Process : Python functions (deposit, withdraw, check balance)

Output : Transaction results

7 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

7.1 Constraints

- Data storage limitations
- Execution speed
- Accuracy of transactions

7.2 Test Plan/ Test Cases

- Test account creation
- Test deposit functionality
- Test withdrawal functionality
- Test balance checking

7.2 Test Procedure

- Run program with different inputs
- Verify outputs
- Check for errors

7.3 Performance Outcome

- System performs operations correctly
- Fast execution
  Accurate results

8 My learnings

During this internship, I learned:

- Python programming concepts
- Functions and control statements
- File handling basics
- Logical thinking and debugging
- Real-world project development

This internship improved my technical skills and prepared me for future job opportunities.

9 Future work scope

- Add database integration (MySQL)
- Develop GUI interface
- Add security features (password, authentication)
- Convert into web application

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